

International Institute of Information Technology, Hyderabad

SC1.440 Dynamical Processes in Complex Networks

Opinion Network Formation

DPCN Assignment 1

Team 34

Kushal Balabhadruni

Rohit Jeswanth

Vishak Kashyap K

GitHub:

<https://github.com/vishakkashyapk30/dpcn-a1-opinion-network-team34>

Table of Contents

1	Team Name	1
2	GitHub Link	1
3	Dataset Documentation	1
3.1	Survey structure	1
3.2	Encoding and filters	1
3.3	Univariate picture	1
4	Pipeline Followed	2
4.1	Conceptual goal	2
4.2	Agent similarity	2
4.3	Threshold selection	2
4.4	Attitude projection	2
5	Analysis and Visualizations	2
5.1	Agent communities (Louvain)	3
5.2	Girvan–Newman cut	4
5.3	Attitude co-endorsement network	5
5.4	Contested vs. community-organizing items	7
6	Results and Discussion	7
6.1	What the network view adds	7
7	Individual Contribution	8
8	References	8

1 Team Name

Team 34

2 GitHub Link

<https://github.com/vishakkashyap30/dpcn-a1-opinion-network-team34>

Reproduction: `code/01_data_and_network.ipynb`, `code/02_analysis.ipynb`, and `code/regenerate_figures.ipynb`.
Figures used below come from that regeneration notebook and from `figures/cytoscape/`.

3 Dataset Documentation

3.1 Survey structure

The survey records opinions from **96** respondents on **60** five-point Likert items, in four blocks of 15: Technology (T), Education (E), Ethics/Society (S), and Environment (V). Following MacCarron et al. (2020), we treat the cleaned table as a **bipartite** participant–item incidence structure. Person–person and statement–statement graphs are projections of that bipartite view, not contact graphs.

3.2 Encoding and filters

Response labels map to $[-1, -0.5, 0, 0.5, 1]$. Blanks and No Comments are missing (not Neutral). Respondents with under **50%** item completeness are dropped, leaving **87** of 96. Pairwise agreement scores require at least 20 jointly answered items.

3.3 Univariate picture

Figure 1 shows a strong Agree skew: mass concentrates at 0.5 and 1.0, with few strong negatives. Domain violins confirm the same lean in every theme, but not equally. Environment is the most cohesive positive block (mean 0.69, variance 0.14); Society follows (mean 0.64). Technology is middling; Education is softest and noisiest (mean 0.45, variance 0.33). Averages alone say “the class mostly agrees.” The network pipeline asks a sharper question: among agreeable respondents, who still clusters with whom, and which items create those clusters?

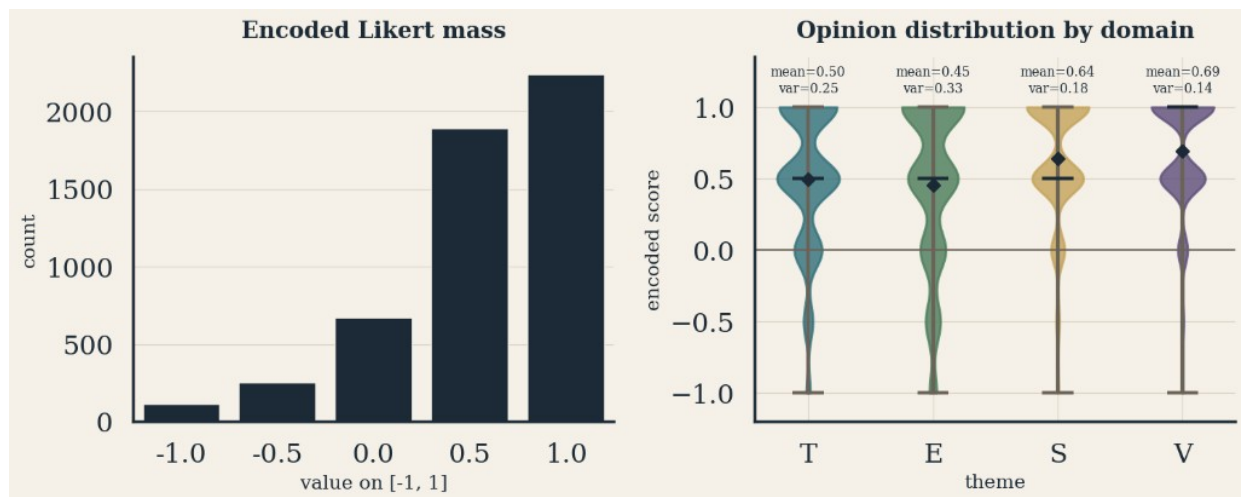


Figure 1: Encoded Likert mass (left) and opinion distributions by theme (right).

4 Pipeline Followed

4.1 Conceptual goal

Following MacCarron et al. (2020), Dinkelberg et al. (2021), and Maher et al. (2020), we build two complementary graphs: an **agent agreement network** (nodes = respondents; edges = high answer similarity) and an **attitude co-endorsement network** (nodes = statements; edges = joint positive endorsement). These are homophily / belief networks.

4.2 Agent similarity

For jointly answered items F_{ij} with $n_{ij} = |F_{ij}|$,

$$S_{ij} = n_{ij} - \sum_{f \in F_{ij}} |q_{if} - q_{jf}|.$$

Larger S_{ij} means closer answer profiles. The empirical S_{ij} distribution (Figure 2, left) peaks near 35 and is left-skewed; most pairs are only moderately similar.

4.3 Threshold selection

We retain edges with $S_{ij} \geq \theta$ and choose the largest θ such that the giant component still covers about **80%** of respondents (Dinkelberg et al., 2021). Figure 2 (right) shows that the giant fraction stays near 1 until roughly $\theta \approx 20$, then declines; it crosses the 0.8 target at $\theta = 41.5$. That cut sits in the upper tail of S_{ij} , so the kept edges are the strongest agreements only. The resulting agent graph has 87 nodes, 456 edges, density 0.122, average degree ≈ 10.5 , and clustering 0.45.

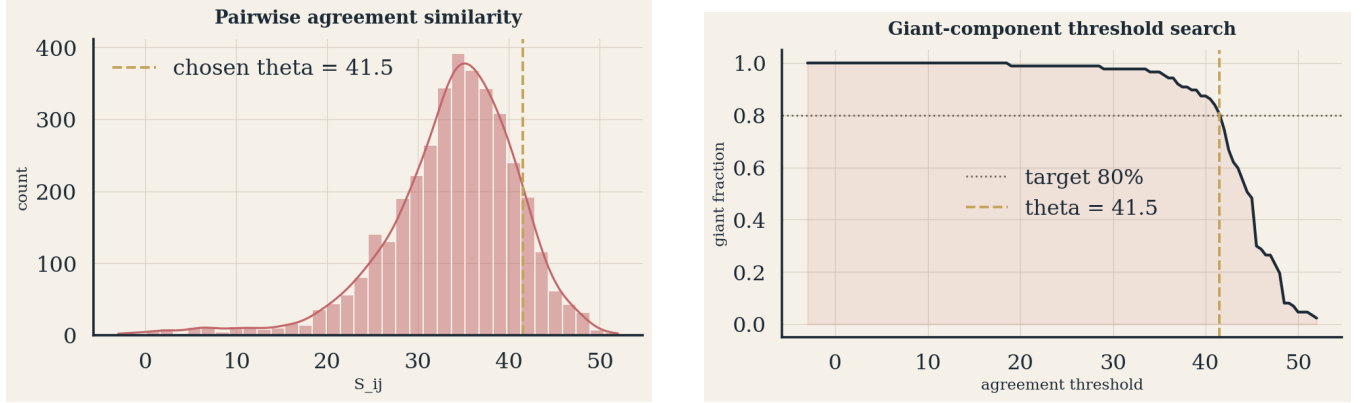


Figure 2: Left: pairwise agreement S_{ij} with chosen $\theta = 41.5$. Right: giant-component fraction vs. threshold (target 80%).

4.4 Attitude projection

Statement pairs use co-endorsement $W_{ab} = \sum_i q_{ia}q_{ib}$ on the $[-1, 1]$ encoding. Strong positive W_{ab} means the two items are routinely affirmed together. We keep high- W edges for visualization and report the top twenty pairs in Table 1.

5 Analysis and Visualizations

Network drawings are Cytoscape.js force-directed views (**cose** layout; edge weight controls spring strength). Interactive HTML lives under `figures/cytoscape/`.

5.1 Agent communities (Louvain)

Louvain on the thresholded agent graph yields moderate modularity ($Q \approx 0.275$). Meaningful mass sits in communities of sizes **37**, **29**, and **4**; remaining labels are mostly isolates. Figure 3 shows a shared core with two large adjacent blocs (C0, C1) and a small peripheral pocket (C2), plus many unconnected respondents outside the giant component.

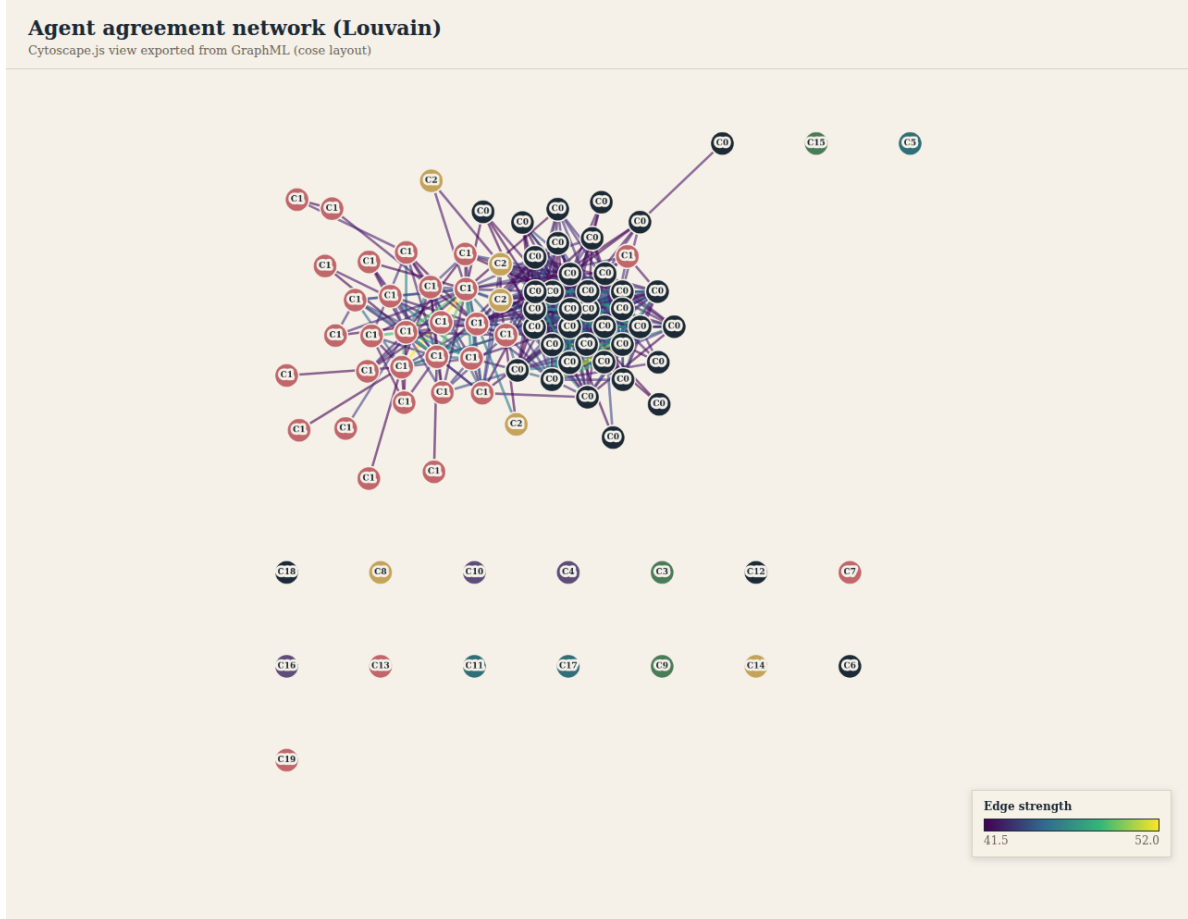


Figure 3: Agent agreement network colored by Louvain community (Cytoscape.js, force-directed by edge weight).

Theme means by community (Figure 4) clarify what the cut captures. Community C0 is not an opposing ideology; it is a **high-intensity** affirmative bloc, especially on Society and Environment (means ≈ 0.82 and 0.87). Community C1 is milder across all themes (means roughly 0.42 – 0.56). Community C2 ($n = 4$) mixes moderate T/E/S with high V. Education remains the lowest theme inside every community. The split is graded affirmation, not left–right issue opposition.

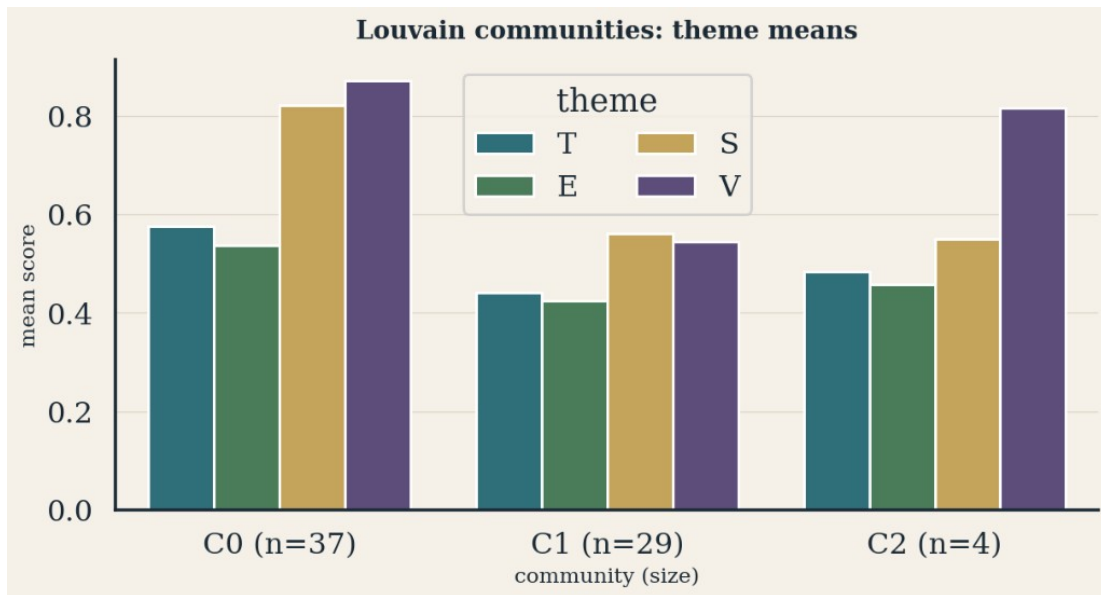


Figure 4: Mean theme scores for Louvain communities with size at least 3.

5.2 Girvan–Newman cut

On the giant component, Girvan–Newman recovers one dominant community and small splinters (Figure 5), rather than two equal camps. Strongest edges (yellow) concentrate inside the red core. Relative to clearer bipolar ANES-style splits in the cited literature, this class looks **weakly polarized**: a dense majority neighborhood with thin fringe groups.

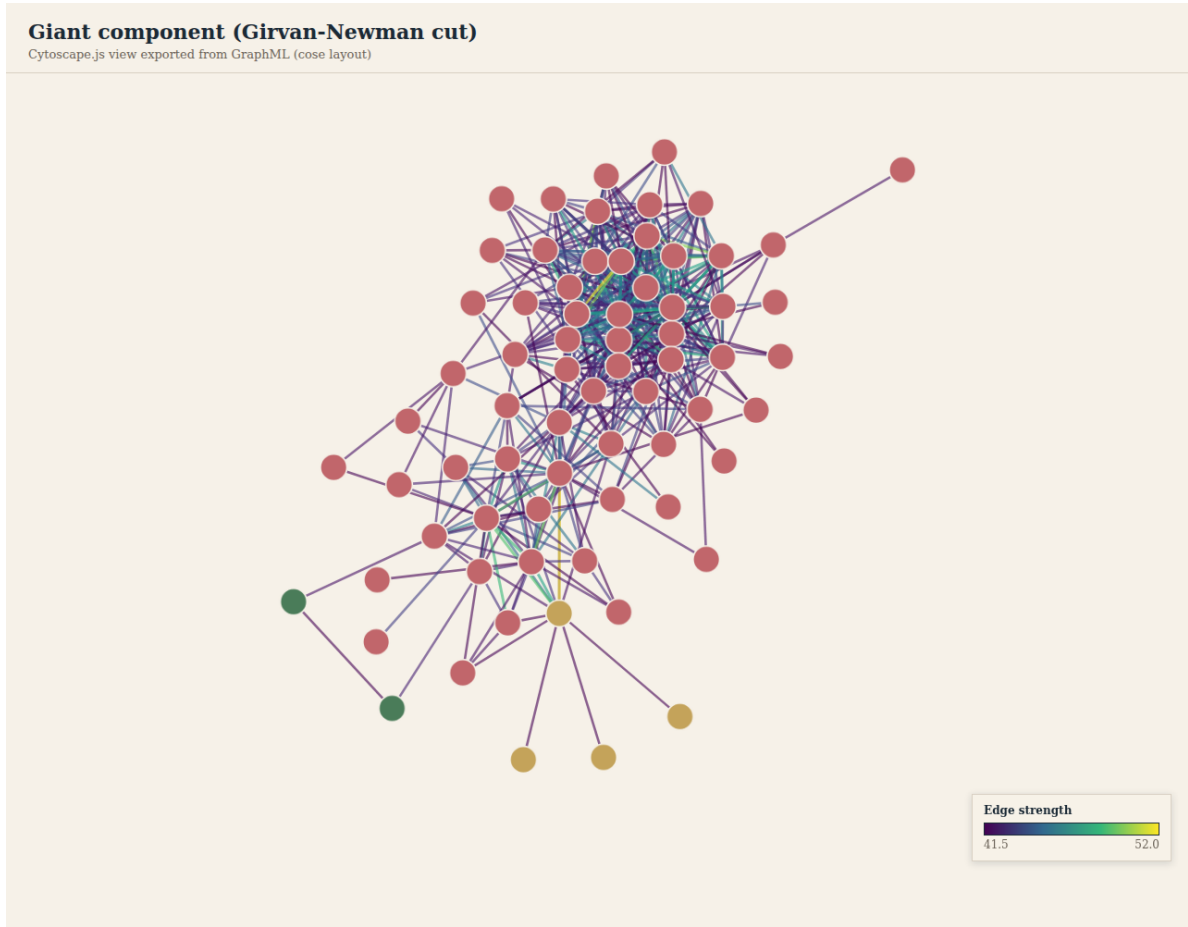


Figure 5: Giant component colored by the Girvan–Newman cut (Cytoscape.js).

5.3 Attitude co-endorsement network

Figure 6 answers which beliefs travel together. Society and Environment items dominate a dense central ball; several Education and Technology items fall out as isolates after thresholding (e.g. E02–E04, E09, E12, T02, T08, T09). Cross-theme bridges exist, but the visual core is an S/V package with selected Education items that everyone affirms (continuous learning, curriculum update), not a four-theme mixture of equal weight.

Attitude co-endorsement network by theme

Cytoscape.js view exported from GraphML (cose layout)

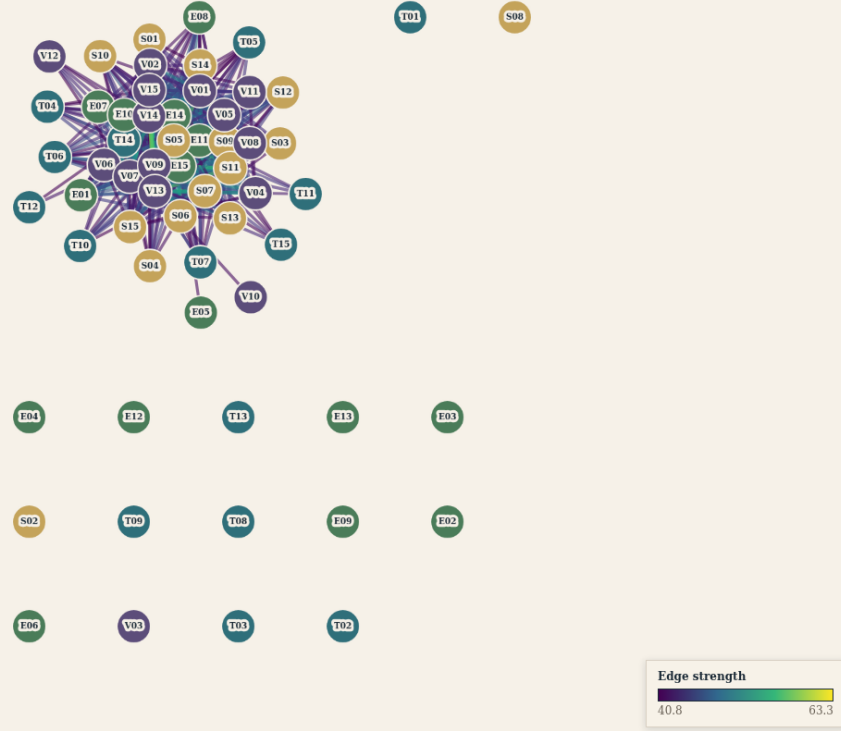


Figure 6: Attitude co-endorsement network colored by theme (Cytoscape.js).

Table 1 lists the twenty strongest co-endorsement pairs. Hub items are **E15** (continuous learning), **E11** (curriculum update), **V13** (corporate environmental accountability), **V09** (reuse/repair design), and **S05** (personal-data control). The top ranks mix within-theme Education pairs (E11–E15, E14–E15) with cross-theme links that glue Education consensus items to the Society/Environment package (E15–V13, E15–S05, S05–V13). That pattern matches the Cytoscape core: broadly shared progressive statements co-travel; contested pedagogy items do not appear in this top list.

Table 1: Top 20 attitude co-endorsement edges (W_{ab}).

#	A	B	Short A	Short B	W
1	E11	E15	Curricula updated for emerging tech	Continuous learning essential	63.25
2	E15	V13	Continuous learning essential	Firms accountable for env. impact	62.00
3	E15	S05	Continuous learning essential	Control over personal data	61.75
4	E14	E15	University–industry collaboration	Continuous learning essential	61.50
5	V09	V13	Design for reuse/repair	Firms accountable for env. impact	61.25
6	E15	V09	Continuous learning essential	Design for reuse/repair	61.00
7	S05	V13	Control over personal data	Firms accountable for env. impact	60.50
8	E10	E15	Innovation over rote learning	Continuous learning essential	60.50
9	E15	S11	Continuous learning essential	Free expression (non-harmful)	60.00
10	E15	V14	Continuous learning essential	Intl. env. cooperation	59.50
11	E11	S05	Curricula updated for emerging tech	Control over personal data	59.50
12	E11	V09	Curricula updated for emerging tech	Design for reuse/repair	59.50
13	V13	V14	Firms accountable for env. impact	Intl. env. cooperation	59.00
14	E11	V13	Curricula updated for emerging tech	Firms accountable for env. impact	59.00
15	S05	V09	Control over personal data	Design for reuse/repair	58.50
16	E10	V13	Innovation over rote learning	Firms accountable for env. impact	58.25
17	V09	V14	Design for reuse/repair	Intl. env. cooperation	58.00
18	E11	S11	Curricula updated for emerging tech	Free expression (non-harmful)	57.75
19	E15	V15	Continuous learning essential	Protect resources for future gens	57.50
20	E10	V09	Innovation over rote learning	Design for reuse/repair	57.25

5.4 Contested vs. community-organizing items

Variance marks disagreement *content*; shuffle importance ($1 - \overline{\text{ARI}}$ after permuting an item vs. baseline Louvain) marks disagreement that *organizes people* (Figure 7).

The most contested items are Education-led: **E04**, **E03**, **E02**, then Technology item T08 and Society item S02. No Environment item enters the top variance list, consistent with V’s tight violin in Figure 1. Shuffle importance also elevates Education (**E04**, **E01**) and mixes in T02, S15, T14, and several V items. So Environment can still structure communities through *intensity* differences (C0 vs. C1) even when raw variance is low, while Education items are both noisy and structurally decisive.



Figure 7: Left: highest-variance (most contested) items. Right: items whose shuffle most disrupts Louvain communities.

6 Results and Discussion

Putting the pieces together yields a coherent class portrait.

1. **Shared affirmative backbone.** Likert mass and theme violins show broad Agree lean. At $\theta = 41.5$ most retained respondents still sit in one giant component ($\sim 80\%$), glued by high-agreement edges and by Society/Environment co-endorsement (Table 1, Figure 6).
2. **Graded neighborhoods, not bipolar camps.** Louvain finds a high-intensity bloc (C0) and a milder bloc (C1); Girvan–Newman finds one large core plus thin splinters. This is weak polarization by affirmation strength, not issue opposition.
3. **Education is the fault line.** Education has the lowest mean and highest domain variance; its items dominate the contested list and top the importance ranking (E04, E01). Contested pedagogy does not appear among the strongest co-endorsement hubs, which instead feature near-consensus Education statements (E15, E11).
4. **Belief packaging.** The attitude network and top- W pairs show a sustainability / data-rights / continuous-learning package. Technology and contested Education items sit on the periphery or as isolates after thresholding.

6.1 What the network view adds

Theme means alone miss backbone-versus-fringe geometry, the two intensity neighborhoods, the contested-versus-organizing item distinction, and co-endorsement packages among statements. The MacCarron/Dinkelberg pipeline turns the Likert table into structure we can draw, threshold, and stress-test: students share a societal–environmental core, while education practice (and related technology items) organize the remaining disagreements.

A practical reading for the class is therefore not “students disagree about the environment,” but “students largely co-endorse sustainability and data-rights statements, and diverge more when asked how universities and AI should teach.” That distinction is visible only once agreement edges, community profiles, contestation, and co-endorsement hubs are read together.

7 Individual Contribution

Member	Contribution
Kushal Balabhadruni	Likert encoding and missingness rules; 50% completeness filter; EDA and theme distribution figures; Dataset Documentation.
Rohit Jeswanth	Agreement score S_{ij} , threshold scan ($\theta = 41.5$), GraphML exports; Louvain / Girvan–Newman; item-importance and variance rankings; Pipeline section.
Vishak Kashyap K	Attitude co-endorsement network and top-20 table; Cytoscape force-directed views; LaTeX report assembly; repository layout and README.

8 References

- MacCarron, P., Maher, P. J., and Quayle, M. (2020). Identifying opinion-based groups from survey data: a bipartite network approach. arXiv:2012.11392.
- Dinkelberg, A., MacCarron, P., Maher, P. J., and Quayle, M. (2021). Detect opinion-based groups and reveal polarisation in survey data. arXiv:2104.14427.
- Maher, P. J., MacCarron, P., and Quayle, M. (2020). Mapping public health responses with attitude networks. *British Journal of Social Psychology*.